One Divisor, Many Incarnations

A Computation-First Map on $\mathbb{P}^1$

A Wall of Computation

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A concrete journey from $D = 2[1] - [2]$ to $L(D)$, $\mathcal{O}(D)$, Cartier local equations, transition functions, global sections, and a projective morphism.

1 Opening Philosophy

A divisor is not a passive piece of bookkeeping. It is a verdict. Written as a finite formal sum of points, D silently legislates the analytic life of every function on the curve: here a pole of bounded order is permitted*, there a zero is* demanded*, and everywhere else nothing irregular may occur. To read a divisor is therefore to read a* ***ledger of allowed poles and forced zeros****. On $\mathbb{P}^1$ this ledger refuses to remain formal. It crystallizes, in turn, into a finite-dimensional vector space $L(D)$, an invertible sheaf $\mathcal{O}(D)$, a list of Cartier local equations, a cocycle of transition functions, a holomorphic line bundle, a space of global sections $H^0(\mathbb{P}^1, \mathcal{O}(D))$, an explicit basis, and finally a morphism to projective space. This page is not a guided tour seen from a safe distance. It is a wall of computation: we pin down a single divisor, $D = 2[1] - [2]$, and force it — by explicit calculation, local data, gluing, and global geometry — to confess every one of its incarnations.*

Throughout we work on

$$X = \mathbb{P}^1_{\mathbb{C}}, \qquad D = 2[1] - [2],$$

with affine coordinate z on $\mathbb{A}^1 = \operatorname{Spec} \mathbb{C}[z] \subset \mathbb{P}^1$, and the chart at infinity governed by

$$w = \frac{1}{z}.$$

The order of vanishing of a rational function at ∞ is measured in the coordinate w; a useful bookkeeping rule for $f \in \mathbb{C}(z)$ is $\operatorname{ord}_\infty(f) = \deg(\text{denominator}) - \deg(\text{numerator})$.

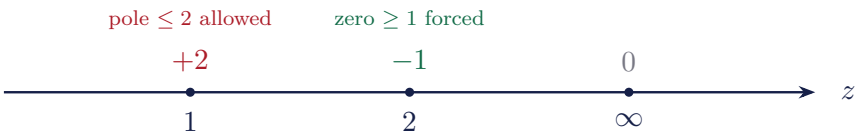
2 The Divisor and Its Meaning

The divisor

$$D = 2[1] - [2]$$

is a rulebook attached to the points of $\mathbb{P}^1$. Reading off its coefficients:

- $2[1]$ — *allows* a pole of order at most 2 at $z = 1$;
- $-[2]$ — *forces* a zero of order at least 1 at $z = 2$;
- no other finite point may host a pole;
- $\deg D = 2 + (-1) = 1$.



Because $\deg D = 1 > 0$, the divisor should be linearly equivalent to a single point. We prove this by exhibiting an explicit principal divisor.

Computation 1 (Linear equivalence $D \sim [\infty]$)**.** Let

$$h(z) = \frac{(z-1)^2}{z-2}.$$

We compute $\operatorname{div}(h)$ point by point.

- At $z = 1$: the numerator vanishes to order 2, nothing cancels, so $\operatorname{ord}_1(h) = +2$.
- At $z = 2$: the denominator vanishes to order 1, so h has a simple pole and $\operatorname{ord}_2(h) = -1$.
- At $z = \infty$: substitute $z = 1/w$,

$$h = \frac{(1/w-1)^2}{1/w-2} = \frac{(1-w)^2/w^2}{(1-2w)/w} = \frac{(1-w)^2}{w\,(1-2w)}.$$

  Near $w = 0$ the numerator tends to 1 while the denominator has a simple zero, so h has a simple pole and $\operatorname{ord}_\infty(h) = -1$.

Hence

$$\operatorname{div}(h) = 2[1] - [2] - [\infty],$$

a degree-zero (principal) divisor, as it must be. Rearranging,

$$D - [\infty] = 2[1] - [2] - [\infty] = \operatorname{div}(h),$$

so

$$\boxed{D \sim [\infty]}.$$

Consequently

$$\mathcal{O}(D) \;\cong\; \mathcal{O}([\infty]) \;\cong\; \mathcal{O}(1).$$

Since $h^0(\mathbb{P}^1, \mathcal{O}(d)) = d + 1$ for $d \geq 0$, this already *predicts* the answer of the next section before we compute it:

$$\dim_{\mathbb{C}} L(D) = h^0(\mathbb{P}^1, \mathcal{O}(1)) = 2.$$

3 Compute $L(D)$ Completely

Definition 1.

$$L(D) = \{\, f \in \mathbb{C}(z) \;:\; \operatorname{div}(f) + D \geq 0 \,\}.$$

For $D = 2[1] - [2]$ the single inequality $\operatorname{div}(f) + D \geq 0$ unpacks, point by point, into:

$$\operatorname{ord}_1(f) \geq -2, \qquad \operatorname{ord}_2(f) \geq +1,$$

$$\operatorname{ord}_p(f) \geq 0 \quad \text{for every finite } p \neq 1, 2, \qquad \operatorname{ord}_\infty(f) \geq 0.$$

In words: f may carry *at most a double pole* at $z = 1$, it *must vanish at least once* at $z = 2$, and it may have *no other pole anywhere*, the point ∞ included.

Deriving the general form. Write $f = P/Q$ in lowest terms.

- The only pole permitted is at $z = 1$ of order ≤ 2, so the denominator is $Q = (z-1)^2$ (taking the largest allowed pole; smaller ones are special cases).

- The forced zero at $z = 2$ means $(z - 2) \mid P$.
- Holomorphy at ∞ requires $\operatorname{ord}_\infty(f) \geq 0$, i.e. $\deg P \leq \deg Q = 2$.

Therefore $P = (z - 2)(az + b)$ with $a, b \in \mathbb{C}$, and

$$\boxed{f(z) = \frac{(z-2)(az+b)}{(z-1)^2}}.$$

The numerator has degree at most 2, matching the denominator's degree 2; hence f tends to the finite value a as $z \to \infty$ and has *no* pole at infinity. The two free parameters a, b confirm $\dim_{\mathbb{C}} L(D) = 2$, exactly as predicted.

Choosing the basis $\{az + b\} = \{1,\ z - 1\}$ of linear polynomials gives

$$L(D) = \operatorname{span}_{\mathbb{C}} \left\{ \frac{z-2}{(z-1)^2},\ \frac{z-2}{z-1} \right\}.$$

Set

$$s_0 = \frac{z-2}{(z-1)^2}, \qquad s_1 = \frac{z-2}{z-1}.$$

section	**at $z = 1$**	**at $z = 2$**	**at ∞**	**allowed by D?**
$s_0 = \dfrac{z-2}{(z-1)^2}$	pole, order 2	zero, order 1	zero, order 1	yes
$s_1 = \dfrac{z-2}{z-1}$	pole, order 1	zero, order 1	unit (value 1)	yes

As a sanity check, $\operatorname{div}(s_0) + D = ([2] + [\infty] - 2[1]) + (2[1] - [2]) = [\infty] \geq 0$, and $\operatorname{div}(s_1) + D = ([2] - [1]) + (2[1] - [2]) = [1] \geq 0$. Both lie in $L(D)$.

4 Translate $L(D)$ into $H^0(\mathbb{P}^1, \mathcal{O}(D))$

The divisor sheaf is defined on each open $U \subseteq \mathbb{P}^1$ by

$$\mathcal{O}(D)(U) = \{\, f \in \mathcal{M}(U) \ :\ \operatorname{div}(f) + D|_U \geq 0 \,\},$$

where $\mathcal{M}$ is the sheaf of meromorphic functions. Taking global sections collapses this to exactly the Riemann–Roch space:

$$H^0(\mathbb{P}^1, \mathcal{O}(D)) = \mathcal{O}(D)(\mathbb{P}^1) = L(D).$$

A global section of $\mathcal{O}(D)$ is a meromorphic function whose bad behavior is exactly compensated by the divisor D.

The divisor D is thus a *local rulebook*: at each point it permits certain poles and demands certain zeros, and H^0 collects precisely those global meromorphic functions that obey every clause of the rulebook at once.

5 Cartier Divisor Data

Cover $\mathbb{P}^1$ by the three opens

$$U_0 = \mathbb{P}^1 \setminus \{1, 2\}, \qquad U_1 = \{\, |z - 1| < r \,\}, \qquad U_2 = \{\, |z - 2| < r \,\},$$

with $r < 1$ small enough that U_1 excludes 2 and U_2 excludes 1 (and the disks meet, if at all, only away from both 1 and 2). Choose local equations

$$f_0 = 1, \qquad f_1 = (z - 1)^2, \qquad f_2 = (z - 2)^{-1}.$$

Each f_i cuts out D on its own patch:

$$\operatorname{div}(f_1) = 2[1] \text{ near } 1, \qquad \operatorname{div}(f_2) = -[2] \text{ near } 2, \qquad \operatorname{div}(f_0) = 0 \text{ away from } 1, 2.$$

This is the defining feature of a **Cartier divisor**: the local data are glued compatibly, meaning the ratios are holomorphic and nowhere zero on overlaps,

$$\frac{f_i}{f_j} \in \mathcal{O}^\times(U_i \cap U_j).$$

Local meromorphic equations for the same divisor can differ only by holomorphic units once the special points have been removed from the overlap.

6 From Cartier Data to $\mathcal{O}(D)$

The sheaf $\mathcal{O}(D)$ is locally free of rank one, trivialized on each patch by the inverse of the local equation:

$$\mathcal{O}(D)\big|_{U_i} = f_i^{-1}\,\mathcal{O}_{U_i}, \qquad e_i := f_i^{-1} \text{ the local generator.}$$

On an overlap the two generators are proportional,

$$e_i = f_i^{-1} = \frac{f_j}{f_i}\, f_j^{-1} = \frac{f_j}{f_i}\, e_j,$$

so with the convention

$$g_{ij} = \frac{f_j}{f_i}$$

we read off the transition functions:

$$g_{10} = \frac{f_0}{f_1} = \frac{1}{(z-1)^2}, \qquad g_{20} = \frac{f_0}{f_2} = z - 2,$$

$$g_{12} = \frac{f_2}{f_1} = \frac{(z-2)^{-1}}{(z-1)^2} = \frac{1}{(z-2)(z-1)^2}.$$

Each is a holomorphic unit on its overlap precisely because the offending point has been removed:

- g_{10} is a unit on $U_1 \cap U_0$ since $z = 1 \notin U_0$;
- g_{20} is a unit on $U_2 \cap U_0$ since $z = 2 \notin U_0$;
- g_{12} is a unit on $U_1 \cap U_2$ since $z = 1, 2 \notin U_1 \cap U_2$ (the disks were chosen small and disjoint from each other's center).

They satisfy the cocycle identity $g_{ij}g_{jk} = g_{ik}$; for instance $g_{10}\, g_{02} = \dfrac{f_0}{f_1} \cdot \dfrac{f_2}{f_0} = \dfrac{f_2}{f_1} = g_{12}$.

7 Interpret $\mathcal{O}(D)$ as a Holomorphic Line Bundle

The same data may be read geometrically. Glue the trivial line bundles

$$U_i \times \mathbb{C}$$

along the overlaps by declaring $(z,\, v_j) \in U_j \times \mathbb{C}$ and $(z,\, v_i) \in U_i \times \mathbb{C}$ to be the same point whenever $v_i = g_{ij}\, v_j$. The result is a holomorphic line bundle whose sheaf of holomorphic sections is exactly $\mathcal{O}(D)$:

$\mathcal{O}(D)$ is obtained by gluing the trivial bundles $U_i \times \mathbb{C}$ using the transition functions g_{ij}.

The invertible sheaf and the holomorphic line bundle are two languages for one object.

$$\begin{array}{ccc} \{(U_i, f_i)\} & \xrightarrow{g_{ij}=f_j/f_i} & \{g_{ij}\} \in Z^1(\mathfrak{U}, \mathcal{O}^\times) \\ \Big\downarrow{\scriptstyle f_i^{-1}\mathcal{O}_{U_i}} & & \Big\downarrow{\scriptstyle \text{glue } U_i\times\mathbb{C}} \\ \mathcal{O}(D) & \xrightarrow[\sim]{} & \text{holomorphic line bundle } L_D \xrightarrow{D\sim[\infty]} \mathcal{O}(1) \end{array}$$

Remark 1. The cocycle $\{g_{ij}\}$ defines a class in $H^1(\mathbb{P}^1, \mathcal{O}^\times) = \mathrm{Pic}(\mathbb{P}^1) \cong \mathbb{Z}$, and that integer is $\deg D = 1$. This is the cohomological face of the computation $D \sim [\infty]$: both say

$$\mathcal{O}(D) \cong \mathcal{O}(1),$$

the unique degree-one line bundle on $\mathbb{P}^1$, which is moreover very ample since $\deg D = 1 \geq 2g+1 = 1$ for $g = 0$.

8 Interpret the Basis as Global Sections

A global section $s \in H^0(\mathbb{P}^1, \mathcal{O}(D))$ is represented on the patch U_i by the *holomorphic* function

$$a_i = f_i\, s.$$

We compute the local representatives of s_0 and s_1.

Near $z = 1$ (using $f_1 = (z-1)^2$):

$$f_1 s_0 = (z-1)^2 \cdot \frac{z-2}{(z-1)^2} = z - 2, \qquad f_1 s_1 = (z-1)^2 \cdot \frac{z-2}{z-1} = (z-1)(z-2).$$

Both are holomorphic near $z = 1$; their values there are -1 and 0.

Near $z = 2$ (using $f_2 = (z-2)^{-1}$):

$$f_2 s_0 = (z-2)^{-1} \cdot \frac{z-2}{(z-1)^2} = \frac{1}{(z-1)^2}, \qquad f_2 s_1 = (z-2)^{-1} \cdot \frac{z-2}{z-1} = \frac{1}{z-1}.$$

Both are holomorphic near $z = 2$; their values there are 1 and 1.

Essential warning. The raw meromorphic functions s_0 and s_1 both vanish at $z = 2$, but the corresponding global sections do *not* vanish simultaneously there. Indeed their honest local values at $z = 2$, read in the U_2-trivialization, are $f_2 s_0\big|_2 = 1$ and $f_2 s_1\big|_2 = 1$ — both nonzero. A section of $\mathcal{O}(D)$ must be evaluated only *after* choosing a local trivialization; the naive number $s_i(2) = 0$ is an artifact of the chart $f_0 = 1$, not a property of the section.

9 Compute the Projective Map

The basis (s_0, s_1) of $H^0(\mathbb{P}^1, \mathcal{O}(D))$ determines a map

$$\Phi_D : \mathbb{P}^1 \longrightarrow \mathbb{P}^1, \qquad p \longmapsto [\, s_0(p) : s_1(p) \,],$$

interpreted through local trivializations: at each point one divides by the local generator before forming homogeneous coordinates.

On U_0 ($f_0 = 1$, so the trivialization is the identity):

$$\Phi_D(z) = \left[\frac{z-2}{(z-1)^2} : \frac{z-2}{z-1} \right].$$

For $z \neq 1, 2$ we may clear the common factor $\dfrac{z-2}{z-1}$ and then the factor $(z-1)$:

$$\left[\frac{z-2}{(z-1)^2} : \frac{z-2}{z-1}\right] = \left[\frac{1}{z-1} : 1\right] = [\,1 : z-1\,].$$

At $z = 2$ (use the U_2-trivialization $f_2 = (z-2)^{-1}$):

$$\Phi_D(2) = \big[\, f_2 s_0 : f_2 s_1 \,\big]_{z=2} = \left[\frac{1}{(2-1)^2} : \frac{1}{2-1}\right] = [\,1 : 1\,],$$

matching $[\,1 : z-1\,]\big|_{z=2} = [1 : 1]$. The point $z = 2$ is therefore *not* a base point.

At $z = 1$ (use the U_1-trivialization $f_1 = (z-1)^2$):

$$\Phi_D(1) = \big[\, f_1 s_0 : f_1 s_1 \,\big]_{z=1} = \big[\,(z-2) : (z-1)(z-2)\,\big]_{z=1} = [-1 : 0] = [\,1 : 0\,],$$

matching $[\,1 : z-1\,]\big|_{z=1} = [1 : 0]$.

At $z = \infty$**:** from $\Phi_D(z) = [\,1 : z-1\,] = \big[\frac{1}{z-1} : 1\big]$,

$$\Phi_D(\infty) = \lim_{z\to\infty}\left[\tfrac{1}{z-1} : 1\right] = [\,0 : 1\,].$$

Collecting every chart,

$$\boxed{\Phi_D(z) = [\,1 : z-1\,]}.$$

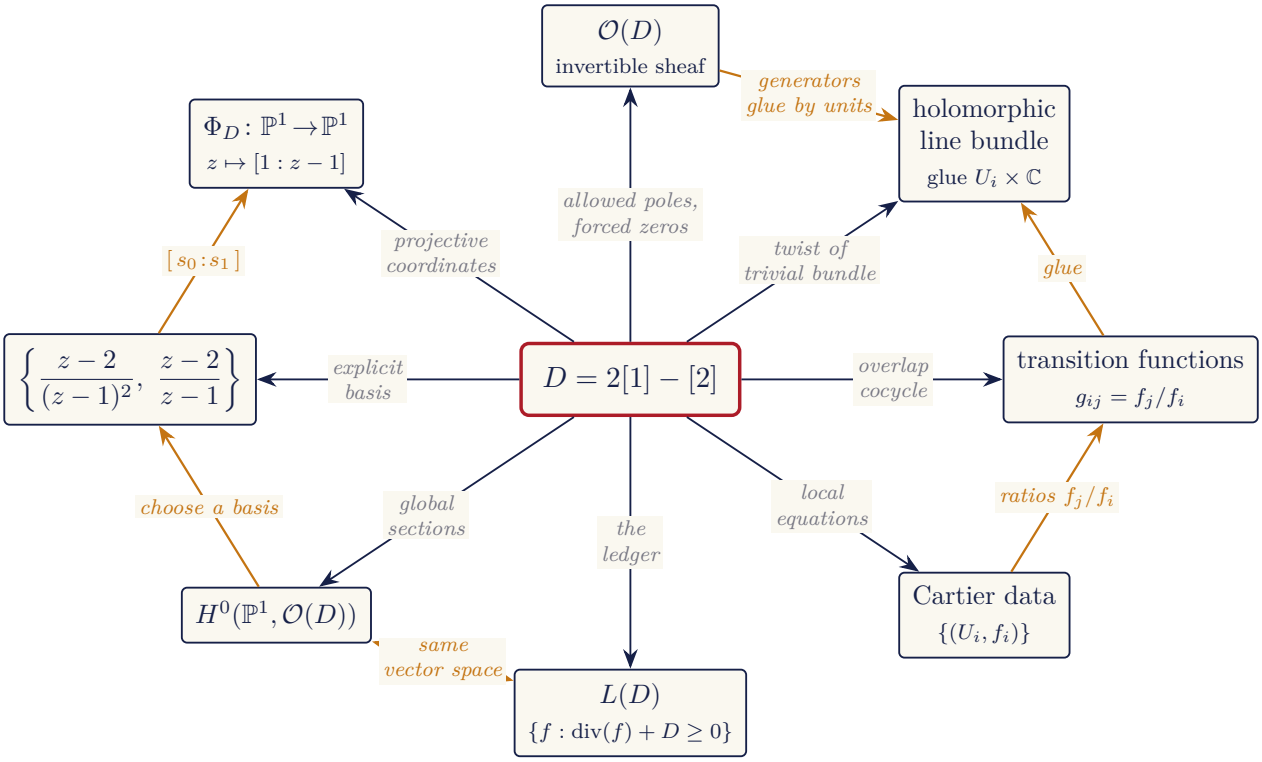
In homogeneous coordinates $z = X/Y$ one has $[\,1 : z-1\,] = [\,1 : \frac{X-Y}{Y}\,] = [\,Y : X-Y\,]$, hence

$$\boxed{[X : Y] \longmapsto [\,Y : X-Y\,]}.$$

The matrix $\left(\begin{smallmatrix} 0 & 1 \\ 1 & -1 \end{smallmatrix}\right)$ has determinant $-1 \neq 0$, so $\Phi_D \in \mathrm{PGL}_2(\mathbb{C})$ is a degree-one *automorphism* of $\mathbb{P}^1$. Equivalently, the complete linear system $|D|$ is base-point free and $\mathcal{O}(D) = \mathcal{O}(1)$ is very ample, so Φ_D is an embedding — here, an isomorphism onto all of $\mathbb{P}^1$.

10 One-Page Conceptual Map

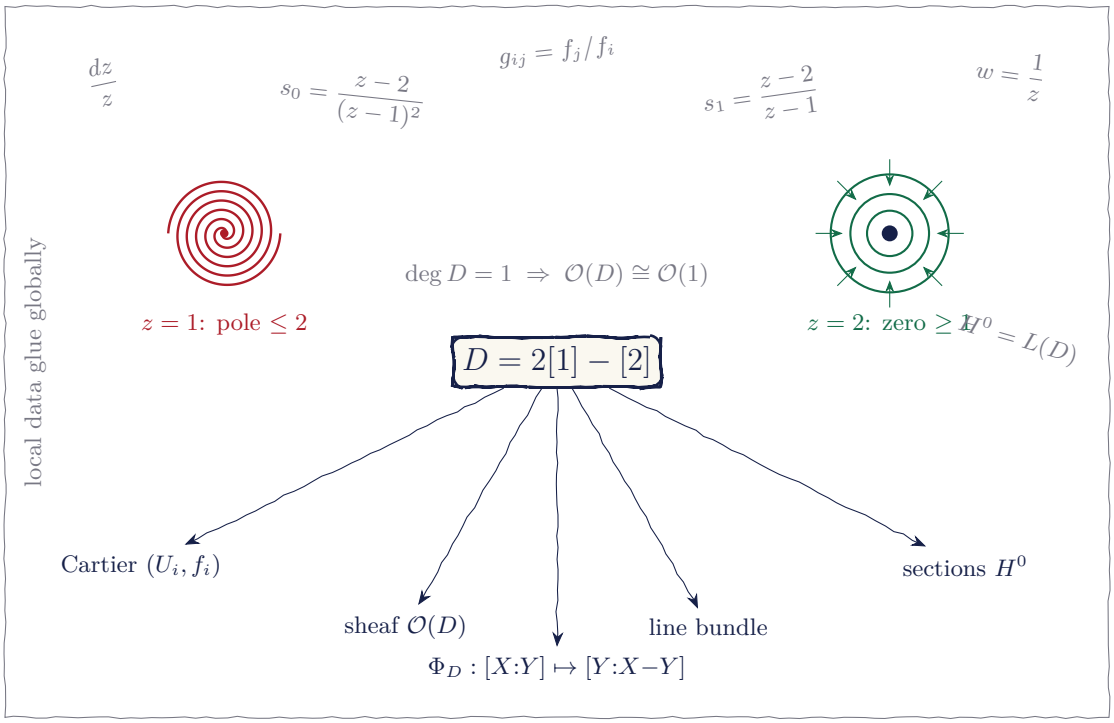
One divisor at the center; eight incarnations on the rim. The blue spokes are the *readings* of D; the amber rim arrows are the *translations* between incarnations.



Sample readings of the labels: $D \to \mathcal{O}(D)$: allowed poles and forced zeros become a sheaf. $\mathcal{O}(D) \to$ line bundle: local generators glue by units. $H^0(\mathcal{O}(D)) \leftrightarrow L(D)$: global sections are meromorphic functions obeying D.

Visual Design: The Divisor as a Wall of Computation

Imagine this page not as clean typeset exposition but as the chalk-and-ink wall of an old mathematician mid-thought. The divisor sits boxed at the center. Around $z = 1$ a **double spiral** coils outward — a pole of order two, two sheets winding away from the point. Around $z = 2$ a **dark sink** swallows the neighborhood, arrows pressing inward toward a forced zero. From the central box, hand-drawn arrows shoot toward the Cartier data, the sheaf, the line bundle, the global sections, and the projective map, while the margins are scrawled with the working formulas. It is not polished exposition, but the wall-writing of someone conquering abstraction by explicit computation.



Final Reflection

So the formal sum $D = 2[1] - [2]$ was never merely formal. Held to the light it is at once a ledger of one forced zero and one bounded pole; a triple of Cartier local equations $(1, (z-1)^2, (z-2)^{-1})$; an invertible sheaf $\mathcal{O}(D)$; a holomorphic line bundle glued from the units $\frac{1}{(z-1)^2}$ and $z - 2$; a two-dimensional space of global meromorphic functions spanned by s_0 and s_1; and the pair of projective coordinates that realize $\mathbb{P}^1$ as itself through $[X : Y] \mapsto [Y : X - Y]$. Abstraction did not have to be believed — it had to be computed. *Local data, gluing, global sections, and projective geometry were all already folded inside three symbols and a difference. That is the quiet violence of a divisor: a single line of notation that, interrogated by calculation, opens into the entire architecture of a curve.*